

KITABU QUEST

Grade 12

AGRICULTURE



Honeybee

Cover scene

learners working in a school garden with seedlings, compost, watering cans, and a friendly honeybee mascot

● Clear lessons

● Worked examples

● Fun activities

● Real-life problems

GRADE

12

Title Page

KITABU QUEST Grade 12 Agriculture

Kenya CBC content draft for review

Mascot: honeybee

This content-first draft was regenerated from Kitabu curriculum data using a topic-by-topic book plan. It is designed to help learners read, practise, discuss, create, and revise with confidence.

Cover artwork is intentionally deferred until all content has passed review and is marked ready-for-cover.

How To Use This Book

This book helps Grade 12 learners study Agriculture through clear lessons, worked examples, activities, practice, and revision.

Use the inquiry questions to think first. Read the explanation. Try the guided example. Complete the activity. Answer the practice questions. Correct your work and ask for support where needed.

Learn through school-garden practice, observation logs, tool safety, and home projects.

Every unit has a local example, a misconception to avoid, success criteria, and a check for understanding.

Learning Skills In This Book

Each chapter builds knowledge, skills, values, creativity, communication, collaboration, critical thinking, digital literacy, and self-confidence.

Look for these page types: Learn, Example, Activity, Practice, Reflection, Home Link, and Review.

How to study well: read the goal first, underline new words, try the example before reading the answer, discuss with a partner, and correct your work after feedback. Do not skip the reflection questions. They help you notice what you understand and what needs more practice.

Table Of Contents

Use this table of contents to plan your study. Start with the first chapter, but return to earlier pages whenever you need revision.

1. 1.0 Crop Production
2. 2.0 Animal Production
3. 3.0 Agricultural Technologies and Entrepreneurship

After each chapter, complete the chapter review before moving forward. At the end of the book, use the glossary, answer notes, and final project to revise the whole subject.

Chapter: Crop Production

In this chapter you will study Crop Production.

Inquiry starter: What do you already know about this topic from home, school, or your community?

Chapter units:

- 1.1 Soil Erosion Control Cut-off drains Terraces Gabions
- 1.2 Ornamental Cropping Forms of ornamental cropping. Constructing ornamental gardens. Importance of ornamental cropping.
- 1.3 Crop Protection: Disease control Categories of crop diseases Crop disease control methods Importance of managing crop
- 1.4 Forage Crops Types of forage Growing forage Ensiling forage

Before you begin, write two things you already know and one question you want to answer. As you study, collect examples from daily life. At the end of the chapter, return to your question and improve your answer using new vocabulary, examples, and evidence from the lessons.

Soil Erosion Control Cut-off drains Terraces Gabions: Lesson Opener

Learning area: Soil Erosion Control Cut-off drains Terraces Gabions

Start here: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Soil Erosion Control Cut-off drains Terraces Gabions.

Curriculum link: This lesson follows the Grade 12 curriculum sequence and focuses on Soil Erosion Control Cut-off drains Terraces Gabions.

By the end of this lesson sequence, you should be able to:

- describe structural measures of controlling soil erosion
- construct structures for soil erosion control
- structural measures in soil erosion control

Inquiry questions:

- How do structural measures control soil erosion?

Before reading, write one thing you already know and one question you want this lesson to answer.

Soil Erosion Control Cut-off drains Terraces Gabions: Learn

Start with this real situation: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Soil Erosion Control Cut-off drains Terraces Gabions.

The important idea is Soil Erosion Control Cut-off drains Terraces Gabions. In this lesson, do not memorise words only. Read the example, try the guided task, then explain the idea in your own words.

A common mistake is describing farm work without naming tools, steps, safety, observations, or care practices. Avoid it by checking your work against these success criteria:

- steps are practical
- tools/materials are named
- safety and care are included

For Grade 12, use exam-ready precision with terms explained before use.

Inline visual support:

| Agriculture record | What to note |

| --- | --- |

| Purpose | Why the task is being done. |

| Tools/materials | Exact items needed for Soil Erosion Control Cut-off drains Terraces Gabions. |

| Safety and care | Hazard, hygiene action, and care for living things. |

| Observation | Date, condition seen, action taken, and result. |

Method for Agriculture: Learn through school-garden practice, observation logs, tool safety, and home projects.

Key words:

- Soil Erosion Control Cut-off drains Terraces Gabions
- Erosion
- Control
- Cut-off
- drains
- Terraces
- Gabions

Explain the idea in your own words before attempting the activities.

Soil Erosion Control Cut-off drains Terraces Gabions: Worked Example

Practical model for Soil Erosion Control Cut-off drains Terraces Gabions: Soil observation and improvement.

Procedure:

1. Collect a small soil sample from an approved school-garden spot.
2. Observe colour, texture, moisture, stones, plant remains, and signs of erosion.
3. Record the evidence in a table before suggesting an improvement.
4. Choose one improvement such as mulching, compost, contour planting, cover crops, or safe drainage.
5. Explain how the improvement protects soil fertility, water, or plant growth.

Hazards: sharp objects, dirty hands, unsafe slopes, and contaminated soil.

Rubric: excellent work names the soil evidence, links it to a practical improvement, includes safety, and records observations clearly.

Soil Erosion Control Cut-off drains Terraces Gabions: Activity

Activity for Soil Erosion Control Cut-off drains Terraces Gabions: Complete a supervised agriculture procedure card.

Inquiry question: How do structural measures control soil erosion?

1. State the purpose of the task.
2. List exact tools, materials, living things, and protective items needed.
3. Write the steps in order, including what the teacher or adult must supervise.
4. Name at least three hazards and the safety action for each hazard.
5. Record what should be observed before, during, and after the task.
6. Judge quality using practical criteria: safety, hygiene, care for living things, correct method, and useful records.

Success criteria:

- steps are practical
- tools/materials are named
- safety and care are included

Home link: Observe a related farm, garden, kitchen, storage, or animal-care practice. Record what was done safely and one improvement.

Soil Erosion Control Cut-off drains Terraces Gabions: Activity And Practice

Practice for Soil Erosion Control Cut-off drains Terraces Gabions:

1. Carry out or describe a practical task for this outcome: describe structural measures of controlling soil erosion. List materials, steps, safety checks, and how you will judge quality.
2. Carry out or describe a practical task for this outcome: construct structures for soil erosion control. List materials, steps, safety checks, and how you will judge quality.
3. Carry out or describe a practical task for this outcome: structural measures in soil erosion control. List materials, steps, safety checks, and how you will judge quality.

Correction check:

- steps are practical
- tools/materials are named
- safety and care are included

Reflection: What was easy? What needs more practice? What question will you ask your teacher or study partner?

Home link: Ask someone at home for an example related to soil erosion control cut-off drains terraces gabions and write two useful sentences about it.

Soil Erosion Control Cut-off drains Terraces Gabions: Outcome Check

Outcome: describe structural measures of controlling soil erosion

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Soil Erosion Control Cut-off drains Terraces Gabions.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Soil Erosion Control Cut-off drains Terraces Gabions: Outcome Check

Outcome: construct structures for soil erosion control

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Soil Erosion Control Cut-off drains Terraces Gabions.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Soil Erosion Control Cut-off drains Terraces Gabions: Outcome Check

Outcome: structural measures in soil erosion control

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Soil Erosion Control Cut-off drains Terraces Gabions.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Ornamental Cropping Forms of ornamental cropping. Constructing ornamental gardens. Importance of ornamental: Lesson Opener

Learning area: Ornamental Cropping Forms of ornamental cropping. Constructing ornamental gardens. Importance of ornamental

Start here: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Ornamental Cropping Forms of ornamental cropping. Constructing ornamental gardens. Importance of ornamental.

Curriculum link: This lesson follows the Grade 12 curriculum sequence and focuses on Ornamental Cropping Forms of ornamental cropping. Constructing ornamental gardens. Importance of ornamental.

By the end of this lesson sequence, you should be able to:

- describe ornamental cropping in environmental aesthetics
- create ornamental garden for aesthetic appeal
- appreciate the role of ornamental cropping as a specialised branch of crop production.

Inquiry questions:

- Why is ornamental cropping important?
- How is ornamental cropping carried out?

Before reading, write one thing you already know and one question you want this lesson to answer.

Ornamental Cropping Forms of ornamental cropping. Constructing ornamental gardens. Importance of ornamental: Learn

Start with this real situation: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Ornamental Cropping Forms of ornamental cropping. Constructing ornamental gardens. Importance of ornamental.

The important idea is Ornamental Cropping Forms of ornamental cropping. Constructing ornamental gardens. Importance of ornamental. In this lesson, do not memorise words only. Read the example, try the guided task, then explain the idea in your own words.

A common mistake is describing farm work without naming tools, steps, safety, observations, or care practices. Avoid it by checking your work against these success criteria:

- steps are practical
- tools/materials are named
- safety and care are included

For Grade 12, use exam-ready precision with terms explained before use.

Inline visual support:

| Agriculture record | What to note |

| --- | --- |

| Purpose | Why the task is being done. |

| Tools/materials | Exact items needed for Ornamental Cropping Forms of ornamental cropping.

Constructing ornamental gardens. Importance of ornamental. |

| Safety and care | Hazard, hygiene action, and care for living things. |

| Observation | Date, condition seen, action taken, and result. |

Method for Agriculture: Learn through school-garden practice, observation logs, tool safety, and home projects.

Key words:

- Ornamental Cropping Forms of ornamental cropping. Constructing ornamental gardens. Importance of ornamental

- Ornamental Cropping Forms of ornamental cropping. Constructing ornamental gardens. Importance of ornamental cropping.

- Ornamental

- Cropping

- Forms

- cropping.

- Constructing

- gardens.

Explain the idea in your own words before attempting the activities.

Ornamental Cropping Forms of ornamental cropping. Constructing ornamental gardens. Importance of ornamental: Worked Example

Practical model for Ornamental Cropping Forms of ornamental cropping. Constructing ornamental gardens. Importance of ornamental: Choose the exact agriculture action, then prove it with evidence.

Procedure:

1. Choose one concrete task from this topic: observe soil, care for a crop, handle a tool, feed or protect an animal, store produce, manage water, or keep a record.
2. State the purpose of that task in one sentence.
3. List exact tools, materials, living things, and safety items before starting.
4. Perform one step at a time under teacher or adult guidance where needed.
5. Record date, condition observed, action taken, result, and one improvement.

Rubric: excellent work names the real task, uses safe handling, gives a clear sequence, cares for living things, and records useful evidence.

Ornamental Cropping Forms of ornamental cropping. Constructing ornamental gardens. Importance of ornamental: Activity

Activity for Ornamental Cropping Forms of ornamental cropping. Constructing ornamental gardens. Importance of ornamental: Complete a supervised agriculture procedure card.

Inquiry question: Why is ornamental cropping important?

1. State the purpose of the task.
2. List exact tools, materials, living things, and protective items needed.
3. Write the steps in order, including what the teacher or adult must supervise.
4. Name at least three hazards and the safety action for each hazard.
5. Record what should be observed before, during, and after the task.
6. Judge quality using practical criteria: safety, hygiene, care for living things, correct method, and useful records.

Success criteria:

- steps are practical
- tools/materials are named
- safety and care are included

Home link: Observe a related farm, garden, kitchen, storage, or animal-care practice. Record what was done safely and one improvement.

Ornamental Cropping Forms of ornamental cropping. Constructing ornamental gardens. Importance of ornamental: Activity And Practice

Practice for Ornamental Cropping Forms of ornamental cropping. Constructing ornamental gardens. Importance of ornamental:

1. Carry out or describe a practical task for this outcome: describe ornamental cropping in environmental aesthetics. List materials, steps, safety checks, and how you will judge quality.
2. Carry out or describe a practical task for this outcome: create ornamental garden for aesthetic appeal. List materials, steps, safety checks, and how you will judge quality.
3. Carry out or describe a practical task for this outcome: appreciate the role of ornamental cropping as a specialised branch of crop production. List materials, steps, safety checks, and how you will judge quality.

Correction check:

- steps are practical
- tools/materials are named
- safety and care are included

Reflection: What was easy? What needs more practice? What question will you ask your teacher or study partner?

Home link: Ask someone at home for an example related to ornamental cropping forms of ornamental cropping. constructing ornamental gardens. importance of ornamental and write two useful sentences about it.

Ornamental Cropping Forms of ornamental cropping. Constructing ornamental gardens. Importance of ornamental: Outcome Check

Outcome: describe ornamental cropping in environmental aesthetics

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Ornamental Cropping Forms of ornamental cropping. Constructing ornamental gardens. Importance of ornamental.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Ornamental Cropping Forms of ornamental cropping. Constructing ornamental gardens. Importance of ornamental: Outcome Check

Outcome: create ornamental garden for aesthetic appeal

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Ornamental Cropping Forms of ornamental cropping. Constructing ornamental gardens. Importance of ornamental.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Ornamental Cropping Forms of ornamental cropping. Constructing ornamental gardens. Importance of ornamental: Outcome Check

Outcome: appreciate the role of ornamental cropping as a specialised branch of crop production.

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Ornamental Cropping Forms of ornamental cropping. Constructing ornamental gardens. Importance of ornamental.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Crop Protection: Disease control Categories of crop diseases

Crop disease control methods: Lesson Opener

Learning area: Crop Protection: Disease control Categories of crop diseases Crop disease control methods

Start here: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Crop Protection: Disease control Categories of crop diseases Crop disease control methods.

Curriculum link: This lesson follows the Grade 12 curriculum sequence and focuses on Crop Protection: Disease control Categories of crop diseases Crop disease control methods.

By the end of this lesson sequence, you should be able to:

- examine categories of diseases in crop production
- evaluate the methods of disease control in crop production
- control diseases in crop production
- managing diseases in crop production

Inquiry questions:

- How do diseases affect crop production?
- How are crop diseases controlled?

Before reading, write one thing you already know and one question you want this lesson to answer.

Crop Protection: Disease control Categories of crop diseases

Crop disease control methods: Learn

Start with this real situation: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Crop Protection: Disease control Categories of crop diseases Crop disease control methods.

The important idea is Crop Protection: Disease control Categories of crop diseases Crop disease control methods. In this lesson, do not memorise words only. Read the example, try the guided task, then explain the idea in your own words.

A common mistake is describing farm work without naming tools, steps, safety, observations, or care practices. Avoid it by checking your work against these success criteria:

- steps are practical
- tools/materials are named
- safety and care are included

For Grade 12, use exam-ready precision with terms explained before use.

Inline visual support:

| Agriculture record | What to note |

| --- | --- |

| Purpose | Why the task is being done. |

| Tools/materials | Exact items needed for Crop Protection: Disease control Categories of crop diseases Crop disease control methods. |

| Safety and care | Hazard, hygiene action, and care for living things. |

| Observation | Date, condition seen, action taken, and result. |

Method for Agriculture: Learn through school-garden practice, observation logs, tool safety, and home projects.

Key words:

- Crop Protection: Disease control Categories of crop diseases Crop disease control methods
- Crop Protection: Disease control Categories of crop diseases Crop disease control methods Importance of managing crop
- Protection
- Disease
- control
- Categories
- diseases
- methods

Explain the idea in your own words before attempting the activities.

Crop Protection: Disease control Categories of crop diseases Crop disease control methods: Worked Example

Practical model for Crop Protection: Disease control Categories of crop diseases Crop disease control methods: Crop-care decision card.

Procedure:

1. Identify the crop stage: seed, nursery, transplanted crop, flowering, fruiting, or harvest.
2. Observe spacing, water, weeds, pests, disease signs, soil cover, and plant colour.
3. Choose one safe action: watering, thinning, weeding, mulching, staking, pest scouting, or harvesting.
4. Record why that action fits the evidence seen.
5. Review the result after a set time and note whether plant health improved.

Hazards: sharp tools, unsafe chemicals, dirty water, plant irritation, and overwatering.

Rubric: excellent work uses real observations, names the crop-care action, protects safety, and explains the expected result.

Crop Protection: Disease control Categories of crop diseases

Crop disease control methods: Activity

Activity for Crop Protection: Disease control Categories of crop diseases Crop disease control methods: Complete a supervised agriculture procedure card.

Inquiry question: How do diseases affect crop production?

1. State the purpose of the task.
2. List exact tools, materials, living things, and protective items needed.
3. Write the steps in order, including what the teacher or adult must supervise.
4. Name at least three hazards and the safety action for each hazard.
5. Record what should be observed before, during, and after the task.
6. Judge quality using practical criteria: safety, hygiene, care for living things, correct method, and useful records.

Success criteria:

- steps are practical
- tools/materials are named
- safety and care are included

Home link: Observe a related farm, garden, kitchen, storage, or animal-care practice. Record what was done safely and one improvement.

Crop Protection: Disease control Categories of crop diseases

Crop disease control methods: Activity And Practice

Practice for Crop Protection: Disease control Categories of crop diseases Crop disease control methods:

1. Carry out or describe a practical task for this outcome: examine categories of diseases in crop production. List materials, steps, safety checks, and how you will judge quality.
2. Carry out or describe a practical task for this outcome: evaluate the methods of disease control in crop production. List materials, steps, safety checks, and how you will judge quality.
3. Carry out or describe a practical task for this outcome: control diseases in crop production. List materials, steps, safety checks, and how you will judge quality.
4. Carry out or describe a practical task for this outcome: managing diseases in crop production. List materials, steps, safety checks, and how you will judge quality.

Correction check:

- steps are practical
- tools/materials are named
- safety and care are included

Reflection: What was easy? What needs more practice? What question will you ask your teacher or study partner?

Home link: Ask someone at home for an example related to crop protection: disease control categories of crop diseases crop disease control methods and write two useful sentences about it.

Crop Protection: Disease control Categories of crop diseases Crop disease control methods: Outcome Check

Outcome: examine categories of diseases in crop production

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Crop Protection: Disease control Categories of crop diseases Crop disease control methods.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Crop Protection: Disease control Categories of crop diseases Crop disease control methods: Outcome Check

Outcome: evaluate the methods of disease control in crop production

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Crop Protection: Disease control Categories of crop diseases Crop disease control methods.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Crop Protection: Disease control Categories of crop diseases Crop disease control methods: Outcome Check

Outcome: control diseases in crop production

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Crop Protection: Disease control Categories of crop diseases Crop disease control methods.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Forage Crops Types of forage Growing forage Ensiling forage: types of forage in crop production: Lesson Opener

Learning area: Forage Crops Types of forage Growing forage Ensiling forage: types of forage in crop production

Start here: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Forage Crops Types of forage Growing forage Ensiling forage: types of forage in crop production.

Curriculum link: This lesson follows the Grade 12 curriculum sequence and focuses on Forage Crops Types of forage Growing forage Ensiling forage: types of forage in crop production.

By the end of this lesson sequence, you should be able to:

- identify types of forage in crop production
- grow a selected forage crop for livestock feed
- prepare silage for forage conservation
- assess the characteristics of good silage for livestock feed

Inquiry questions:

- How are forage crops established?
- How is forage conserved?

Before reading, write one thing you already know and one question you want this lesson to answer.

Forage Crops Types of forage Growing forage Ensiling forage: types of forage in crop production: Learn

Start with this real situation: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Forage Crops Types of forage Growing forage Ensiling forage: types of forage in crop production.

The important idea is Forage Crops Types of forage Growing forage Ensiling forage: types of forage in crop production. In this lesson, do not memorise words only. Read the example, try the guided task, then explain the idea in your own words.

A common mistake is describing farm work without naming tools, steps, safety, observations, or care practices. Avoid it by checking your work against these success criteria:

- steps are practical
- tools/materials are named
- safety and care are included

For Grade 12, use exam-ready precision with terms explained before use.

Inline visual support:

| Agriculture record | What to note |

| --- | --- |

| Purpose | Why the task is being done. |

| Tools/materials | Exact items needed for Forage Crops Types of forage Growing forage Ensiling forage: types of forage in crop production. |

| Safety and care | Hazard, hygiene action, and care for living things. |

| Observation | Date, condition seen, action taken, and result. |

Method for Agriculture: Learn through school-garden practice, observation logs, tool safety, and home projects.

Key words:

- Forage Crops Types of forage Growing forage Ensiling forage: types of forage in crop production
- Forage Crops Types of forage Growing forage Ensiling forage
- Forage
- Crops
- Types
- Growing
- Ensiling
- production

Explain the idea in your own words before attempting the activities.

Forage Crops Types of forage Growing forage Ensiling forage: types of forage in crop production: Worked Example

Practical model for Forage Crops Types of forage Growing forage Ensiling forage: types of forage in crop production: Crop-care decision card.

Procedure:

1. Identify the crop stage: seed, nursery, transplanted crop, flowering, fruiting, or harvest.
2. Observe spacing, water, weeds, pests, disease signs, soil cover, and plant colour.
3. Choose one safe action: watering, thinning, weeding, mulching, staking, pest scouting, or harvesting.
4. Record why that action fits the evidence seen.
5. Review the result after a set time and note whether plant health improved.

Hazards: sharp tools, unsafe chemicals, dirty water, plant irritation, and overwatering.

Rubric: excellent work uses real observations, names the crop-care action, protects safety, and explains the expected result.

Forage Crops Types of forage Growing forage Ensiling forage: types of forage in crop production: Activity

Activity for Forage Crops Types of forage Growing forage Ensiling forage: types of forage in crop production: Complete a supervised agriculture procedure card.

Inquiry question: How are forage crops established?

1. State the purpose of the task.
2. List exact tools, materials, living things, and protective items needed.
3. Write the steps in order, including what the teacher or adult must supervise.
4. Name at least three hazards and the safety action for each hazard.
5. Record what should be observed before, during, and after the task.
6. Judge quality using practical criteria: safety, hygiene, care for living things, correct method, and useful records.

Success criteria:

- steps are practical
- tools/materials are named
- safety and care are included

Home link: Observe a related farm, garden, kitchen, storage, or animal-care practice. Record what was done safely and one improvement.

Forage Crops Types of forage Growing forage Ensiling forage: types of forage in crop production: Activity And Practice

Practice for Forage Crops Types of forage Growing forage Ensiling forage: types of forage in crop production:

1. Carry out or describe a practical task for this outcome: identify types of forage in crop production. List materials, steps, safety checks, and how you will judge quality.
2. Carry out or describe a practical task for this outcome: grow a selected forage crop for livestock feed. List materials, steps, safety checks, and how you will judge quality.
3. Carry out or describe a practical task for this outcome: prepare silage for forage conservation. List materials, steps, safety checks, and how you will judge quality.
4. Carry out or describe a practical task for this outcome: assess the characteristics of good silage for livestock feed. List materials, steps, safety checks, and how you will judge quality.

Correction check:

- steps are practical
- tools/materials are named
- safety and care are included

Reflection: What was easy? What needs more practice? What question will you ask your teacher or study partner?

Home link: Ask someone at home for an example related to forage crops types of forage growing forage ensiling forage: types of forage in crop production and write two useful sentences about it.

Forage Crops Types of forage Growing forage Ensiling forage: types of forage in crop production: Outcome Check

Outcome: identify types of forage in crop production

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Forage Crops Types of forage Growing forage Ensiling forage: types of forage in crop production.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Forage Crops Types of forage Growing forage Ensiling forage: types of forage in crop production: Outcome Check

Outcome: grow a selected forage crop for livestock feed

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Forage Crops Types of forage Growing forage Ensiling forage: types of forage in crop production.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Forage Crops Types of forage Growing forage Ensiling forage: types of forage in crop production: Outcome Check

Outcome: prepare silage for forage conservation

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Forage Crops Types of forage Growing forage Ensiling forage: types of forage in crop production.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Forage Crops Types of forage Growing forage Ensiling forage: acknowledge the importance of growing forage crops: Lesson Opener

Learning area: Forage Crops Types of forage Growing forage Ensiling forage: acknowledge the importance of growing forage crops

Start here: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Forage Crops Types of forage Growing forage Ensiling forage: acknowledge the importance of growing forage crops.

Curriculum link: This lesson follows the Grade 12 curriculum sequence and focuses on Forage Crops Types of forage Growing forage Ensiling forage: acknowledge the importance of growing forage crops.

By the end of this lesson sequence, you should be able to:

- acknowledge the importance of growing forage crops in agricultural production.

Inquiry questions:

- How is forage conserved?

Before reading, write one thing you already know and one question you want this lesson to answer.

Forage Crops Types of forage Growing forage Ensiling forage: acknowledge the importance of growing forage crops: Learn

Start with this real situation: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Forage Crops Types of forage Growing forage Ensiling forage: acknowledge the importance of growing forage crops.

The important idea is Forage Crops Types of forage Growing forage Ensiling forage: acknowledge the importance of growing forage crops. In this lesson, do not memorise words only. Read the example, try the guided task, then explain the idea in your own words.

A common mistake is describing farm work without naming tools, steps, safety, observations, or care practices. Avoid it by checking your work against these success criteria:

- steps are practical
- tools/materials are named
- safety and care are included

For Grade 12, use exam-ready precision with terms explained before use.

Inline visual support:

| Agriculture record | What to note |

| --- | --- |

| Purpose | Why the task is being done. |

| Tools/materials | Exact items needed for Forage Crops Types of forage Growing forage Ensiling forage: acknowledge the importance of growing forage crops. |

| Safety and care | Hazard, hygiene action, and care for living things. |

| Observation | Date, condition seen, action taken, and result. |

Method for Agriculture: Learn through school-garden practice, observation logs, tool safety, and home projects.

Key words:

- Forage Crops Types of forage Growing forage Ensiling forage: acknowledge the importance of growing forage crops
- Forage Crops Types of forage Growing forage Ensiling forage
- Forage
- Crops
- Types
- Growing
- Ensiling
- acknowledge

Explain the idea in your own words before attempting the activities.

Forage Crops Types of forage Growing forage Ensiling forage: acknowledge the importance of growing forage crops: Worked Example

Practical model for Forage Crops Types of forage Growing forage Ensiling forage: acknowledge the importance of growing forage crops: Choose the exact agriculture action, then prove it with evidence.

Procedure:

1. Choose one concrete task from this topic: observe soil, care for a crop, handle a tool, feed or protect an animal, store produce, manage water, or keep a record.
2. State the purpose of that task in one sentence.
3. List exact tools, materials, living things, and safety items before starting.
4. Perform one step at a time under teacher or adult guidance where needed.
5. Record date, condition observed, action taken, result, and one improvement.

Rubric: excellent work names the real task, uses safe handling, gives a clear sequence, cares for living things, and records useful evidence.

Forage Crops Types of forage Growing forage Ensiling forage: acknowledge the importance of growing forage crops: Activity

Activity for Forage Crops Types of forage Growing forage Ensiling forage: acknowledge the importance of growing forage crops: Complete a supervised agriculture procedure card.

Inquiry question: How is forage conserved?

1. State the purpose of the task.
2. List exact tools, materials, living things, and protective items needed.
3. Write the steps in order, including what the teacher or adult must supervise.
4. Name at least three hazards and the safety action for each hazard.
5. Record what should be observed before, during, and after the task.
6. Judge quality using practical criteria: safety, hygiene, care for living things, correct method, and useful records.

Success criteria:

- steps are practical
- tools/materials are named
- safety and care are included

Home link: Observe a related farm, garden, kitchen, storage, or animal-care practice. Record what was done safely and one improvement.

Forage Crops Types of forage Growing forage Ensiling forage: acknowledge the importance of growing forage crops: Activity And Practice

Practice for Forage Crops Types of forage Growing forage Ensiling forage: acknowledge the importance of growing forage crops:

1. Carry out or describe a practical task for this outcome: acknowledge the importance of growing forage crops in agricultural production. List materials, steps, safety checks, and how you will judge quality.

Correction check:

- steps are practical
- tools/materials are named
- safety and care are included

Reflection: What was easy? What needs more practice? What question will you ask your teacher or study partner?

Home link: Ask someone at home for an example related to forage crops types of forage growing forage ensiling forage: acknowledge the importance of growing forage crops and write two useful sentences about it.

Forage Crops Types of forage Growing forage Ensiling forage: acknowledge the importance of growing forage crops: Outcome Check

Outcome: acknowledge the importance of growing forage crops in agricultural production.

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Forage Crops Types of forage Growing forage Ensiling forage: acknowledge the importance of growing forage crops.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Crop Production: Chapter Review

Review the chapter carefully.

1. Write five key words from this chapter and explain each one.
2. Choose two units and compare what you learned.
3. Create one poster, chart, dialogue, model, map, or worked solution that teaches this chapter.
4. Answer one inquiry question again. Improve your first answer using what you now know.
5. Rate your confidence: high, medium, or needs practice.

Chapter: Animal Production

In this chapter you will study Animal Production.

Inquiry starter: What do you already know about this topic from home, school, or your community?

Chapter units:

- 2.1 Livestock Nutrition Role of nutrients Roughages and concentrates Computation of feed ration Preparation of feed ration.
- 2.2 Animal Breeding Reproductive systems Reproductive stages Breeding systems Selection and culling Methods of service Role
- 2.3 Animal Health: Disease Control Predisposing factors Categories of animal diseases Disease control measures Importance of disease
- 2.4 Non-conventional Livestock and Companion Animals Non-conventional livestock Companion animals Roles of companion animals and non-conventional livestock.

Before you begin, write two things you already know and one question you want to answer. As you study, collect examples from daily life. At the end of the chapter, return to your question and improve your answer using new vocabulary, examples, and evidence from the lessons.

Livestock Nutrition Role of nutrients Roughages and concentrates Computation of feed ration: explain the role: Lesson Opener

Learning area: Livestock Nutrition Role of nutrients Roughages and concentrates Computation of feed ration: explain the role

Start here: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Livestock Nutrition Role of nutrients Roughages and concentrates Computation of feed ration: explain the role.

Curriculum link: This lesson follows the Grade 12 curriculum sequence and focuses on Livestock Nutrition Role of nutrients Roughages and concentrates Computation of feed ration: explain the role.

By the end of this lesson sequence, you should be able to:

- explain the role of nutrients in animal production
- examine types of feeds for livestock nutrition
- compute a feed ration for selected animals
- prepare a sample feed ration for a selected animal

Inquiry questions:

- How does animal nutrition influence productivity?

Before reading, write one thing you already know and one question you want this lesson to answer.

Livestock Nutrition Role of nutrients Roughages and concentrates Computation of feed ration: explain the role: Learn

Start with this real situation: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Livestock Nutrition Role of nutrients Roughages and concentrates Computation of feed ration: explain the role.

The important idea is Livestock Nutrition Role of nutrients Roughages and concentrates Computation of feed ration: explain the role. In this lesson, do not memorise words only. Read the example, try the guided task, then explain the idea in your own words.

A common mistake is describing farm work without naming tools, steps, safety, observations, or care practices. Avoid it by checking your work against these success criteria:

- steps are practical
- tools/materials are named
- safety and care are included

For Grade 12, use exam-ready precision with terms explained before use.

Inline visual support:

| Agriculture record | What to note |

| --- | --- |

| Purpose | Why the task is being done. |

| Tools/materials | Exact items needed for Livestock Nutrition Role of nutrients Roughages and concentrates Computation of feed ration: explain the role. |

| Safety and care | Hazard, hygiene action, and care for living things. |

| Observation | Date, condition seen, action taken, and result. |

Method for Agriculture: Learn through school-garden practice, observation logs, tool safety, and home projects.

Key words:

- Livestock Nutrition Role of nutrients Roughages and concentrates Computation of feed ration: explain the role

- Livestock Nutrition Role of nutrients Roughages and concentrates Computation of feed ration Preparation of feed ration.

- Livestock

- Nutrition

- nutrients

- Roughages

- concentrates

- Computation

Explain the idea in your own words before attempting the activities.

Livestock Nutrition Role of nutrients Roughages and concentrates Computation of feed ration: explain the role: Worked Example

Practical model for Livestock Nutrition Role of nutrients Roughages and concentrates Computation of feed ration: explain the role: Safe animal handling.

Teacher-supervised procedure:

1. Observe the animal's behaviour before moving close.
2. Approach from the side where the animal can see you.
3. Keep hands away from the mouth, horns, hooves, and back legs.
4. Use clean water, feed, bedding, and equipment as instructed.
5. Wash hands and disinfect equipment after the task.

Hazards: kicks, bites, scratches, disease transmission, and stress to the animal.

Quality criteria: calm handling, clean tools, correct restraint if needed, animal welfare, and accurate records of feed, health, or behaviour.

Livestock Nutrition Role of nutrients Roughages and concentrates Computation of feed ration: explain the role: Activity

Activity for Livestock Nutrition Role of nutrients Roughages and concentrates Computation of feed ration: explain the role: Complete a supervised agriculture procedure card.

Inquiry question: How does animal nutrition influence productivity?

1. State the purpose of the task.
2. List exact tools, materials, living things, and protective items needed.
3. Write the steps in order, including what the teacher or adult must supervise.
4. Name at least three hazards and the safety action for each hazard.
5. Record what should be observed before, during, and after the task.
6. Judge quality using practical criteria: safety, hygiene, care for living things, correct method, and useful records.

Success criteria:

- steps are practical
- tools/materials are named
- safety and care are included

Home link: Observe a related farm, garden, kitchen, storage, or animal-care practice. Record what was done safely and one improvement.

Livestock Nutrition Role of nutrients Roughages and concentrates Computation of feed ration: explain the role: Activity And Practice

Practice for Livestock Nutrition Role of nutrients Roughages and concentrates Computation of feed ration: explain the role:

1. Carry out or describe a practical task for this outcome: explain the role of nutrients in animal production. List materials, steps, safety checks, and how you will judge quality.
2. Carry out or describe a practical task for this outcome: examine types of feeds for livestock nutrition. List materials, steps, safety checks, and how you will judge quality.
3. Carry out or describe a practical task for this outcome: compute a feed ration for selected animals. List materials, steps, safety checks, and how you will judge quality.
4. Carry out or describe a practical task for this outcome: prepare a sample feed ration for a selected animal. List materials, steps, safety checks, and how you will judge quality.

Correction check:

- steps are practical
- tools/materials are named
- safety and care are included

Reflection: What was easy? What needs more practice? What question will you ask your teacher or study partner?

Home link: Ask someone at home for an example related to livestock nutrition role of nutrients roughages and concentrates computation of feed ration: explain the role and write two useful sentences about it.

Livestock Nutrition Role of nutrients Roughages and concentrates Computation of feed ration: explain the role: Outcome Check

Outcome: explain the role of nutrients in animal production

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Livestock Nutrition Role of nutrients Roughages and concentrates Computation of feed ration: explain the role.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Livestock Nutrition Role of nutrients Roughages and concentrates Computation of feed ration: explain the role: Outcome Check

Outcome: examine types of feeds for livestock nutrition

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Livestock Nutrition Role of nutrients Roughages and concentrates Computation of feed ration: explain the role.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Livestock Nutrition Role of nutrients Roughages and concentrates Computation of feed ration: explain the role: Outcome Check

Outcome: compute a feed ration for selected animals

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Livestock Nutrition Role of nutrients Roughages and concentrates Computation of feed ration: explain the role.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Livestock Nutrition Role of nutrients Roughages and concentrates Computation of feed ration: role of nutrition: Lesson Opener

Learning area: Livestock Nutrition Role of nutrients Roughages and concentrates Computation of feed ration: role of nutrition

Start here: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Livestock Nutrition Role of nutrients Roughages and concentrates Computation of feed ration: role of nutrition.

Curriculum link: This lesson follows the Grade 12 curriculum sequence and focuses on Livestock Nutrition Role of nutrients Roughages and concentrates Computation of feed ration: role of nutrition.

By the end of this lesson sequence, you should be able to:

- appreciate the role of nutrition in animal production.

Inquiry questions:

- How does animal nutrition influence productivity?

Before reading, write one thing you already know and one question you want this lesson to answer.

Livestock Nutrition Role of nutrients Roughages and concentrates Computation of feed ration: role of nutrition: Learn

Start with this real situation: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Livestock Nutrition Role of nutrients Roughages and concentrates Computation of feed ration: role of nutrition.

The important idea is Livestock Nutrition Role of nutrients Roughages and concentrates Computation of feed ration: role of nutrition. In this lesson, do not memorise words only. Read the example, try the guided task, then explain the idea in your own words.

A common mistake is describing farm work without naming tools, steps, safety, observations, or care practices. Avoid it by checking your work against these success criteria:

- steps are practical
- tools/materials are named
- safety and care are included

For Grade 12, use exam-ready precision with terms explained before use.

Inline visual support:

| Agriculture record | What to note |

| --- | --- |

| Purpose | Why the task is being done. |

| Tools/materials | Exact items needed for Livestock Nutrition Role of nutrients Roughages and concentrates Computation of feed ration: role of nutrition. |

| Safety and care | Hazard, hygiene action, and care for living things. |

| Observation | Date, condition seen, action taken, and result. |

Method for Agriculture: Learn through school-garden practice, observation logs, tool safety, and home projects.

Key words:

- Livestock Nutrition Role of nutrients Roughages and concentrates Computation of feed ration: role of nutrition

- Livestock Nutrition Role of nutrients Roughages and concentrates Computation of feed ration Preparation of feed ration.

- Livestock

- Nutrition

- nutrients

- Roughages

- concentrates

- Computation

Explain the idea in your own words before attempting the activities.

Livestock Nutrition Role of nutrients Roughages and concentrates Computation of feed ration: role of nutrition: Worked Example

Practical model for Livestock Nutrition Role of nutrients Roughages and concentrates Computation of feed ration: role of nutrition: Safe animal handling.

Teacher-supervised procedure:

1. Observe the animal's behaviour before moving close.
2. Approach from the side where the animal can see you.
3. Keep hands away from the mouth, horns, hooves, and back legs.
4. Use clean water, feed, bedding, and equipment as instructed.
5. Wash hands and disinfect equipment after the task.

Hazards: kicks, bites, scratches, disease transmission, and stress to the animal.

Quality criteria: calm handling, clean tools, correct restraint if needed, animal welfare, and accurate records of feed, health, or behaviour.

Livestock Nutrition Role of nutrients Roughages and concentrates Computation of feed ration: role of nutrition: Activity

Activity for Livestock Nutrition Role of nutrients Roughages and concentrates Computation of feed ration: role of nutrition: Complete a supervised agriculture procedure card.

Inquiry question: How does animal nutrition influence productivity?

1. State the purpose of the task.
2. List exact tools, materials, living things, and protective items needed.
3. Write the steps in order, including what the teacher or adult must supervise.
4. Name at least three hazards and the safety action for each hazard.
5. Record what should be observed before, during, and after the task.
6. Judge quality using practical criteria: safety, hygiene, care for living things, correct method, and useful records.

Success criteria:

- steps are practical
- tools/materials are named
- safety and care are included

Home link: Observe a related farm, garden, kitchen, storage, or animal-care practice. Record what was done safely and one improvement.

Livestock Nutrition Role of nutrients Roughages and concentrates Computation of feed ration: role of nutrition: Activity And Practice

Practice for Livestock Nutrition Role of nutrients Roughages and concentrates Computation of feed ration: role of nutrition:

1. Carry out or describe a practical task for this outcome: appreciate the role of nutrition in animal production. List materials, steps, safety checks, and how you will judge quality.

Correction check:

- steps are practical
- tools/materials are named
- safety and care are included

Reflection: What was easy? What needs more practice? What question will you ask your teacher or study partner?

Home link: Ask someone at home for an example related to livestock nutrition role of nutrients roughages and concentrates computation of feed ration: role of nutrition and write two useful sentences about it.

Livestock Nutrition Role of nutrients Roughages and concentrates Computation of feed ration: role of nutrition: Outcome Check

Outcome: appreciate the role of nutrition in animal production.

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Livestock Nutrition Role of nutrients Roughages and concentrates Computation of feed ration: role of nutrition.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Animal Breeding Reproductive systems Reproductive stages Breeding systems Selection and culling Methods: reproductive systems of animals: Lesson Opener

Learning area: Animal Breeding Reproductive systems Reproductive stages Breeding systems
Selection and culling Methods: reproductive systems of animals

Start here: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security
example linked to Animal Breeding Reproductive systems Reproductive stages Breeding systems
Selection and culling Methods: reproductive systems of animals.

Curriculum link: This lesson follows the Grade 12 curriculum sequence and focuses on Animal
Breeding Reproductive systems Reproductive stages Breeding systems Selection and culling
Methods: reproductive systems of animals.

By the end of this lesson sequence, you should be able to:

- describe reproductive systems of animals
- examine reproductive stages in animal production
- assess breeding systems in animal production
- select breeding stock in animal production

Inquiry questions:

- Why are animals selected for breeding?
- How are animals selected for breeding?

Before reading, write one thing you already know and one question you want this lesson to answer.

Animal Breeding Reproductive systems Reproductive stages Breeding systems Selection and culling Methods: reproductive systems of animals: Learn

Start with this real situation: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Animal Breeding Reproductive systems Reproductive stages Breeding systems Selection and culling Methods: reproductive systems of animals.

The important idea is Animal Breeding Reproductive systems Reproductive stages Breeding systems Selection and culling Methods: reproductive systems of animals. In this lesson, do not memorise words only. Read the example, try the guided task, then explain the idea in your own words.

A common mistake is describing farm work without naming tools, steps, safety, observations, or care practices. Avoid it by checking your work against these success criteria:

- steps are practical
- tools/materials are named
- safety and care are included

For Grade 12, use exam-ready precision with terms explained before use.

Inline visual support:

| Agriculture record | What to note |

| --- | --- |

| Purpose | Why the task is being done. |

| Tools/materials | Exact items needed for Animal Breeding Reproductive systems Reproductive stages Breeding systems Selection and culling Methods: reproductive systems of animals. |

| Safety and care | Hazard, hygiene action, and care for living things. |

| Observation | Date, condition seen, action taken, and result. |

Method for Agriculture: Learn through school-garden practice, observation logs, tool safety, and home projects.

Key words:

- Animal Breeding Reproductive systems Reproductive stages Breeding systems Selection and culling Methods: reproductive systems of animals
- Animal Breeding Reproductive systems Reproductive stages Breeding systems Selection and culling Methods of service Role
- Animal
- Breeding
- Reproductive
- systems
- stages
- Selection

Explain the idea in your own words before attempting the activities.

Animal Breeding Reproductive systems Reproductive stages Breeding systems Selection and culling Methods: reproductive systems of animals: Worked Example

Practical model for Animal Breeding Reproductive systems Reproductive stages Breeding systems Selection and culling Methods: reproductive systems of animals: Choose the exact agriculture action, then prove it with evidence.

Procedure:

1. Choose one concrete task from this topic: observe soil, care for a crop, handle a tool, feed or protect an animal, store produce, manage water, or keep a record.
2. State the purpose of that task in one sentence.
3. List exact tools, materials, living things, and safety items before starting.
4. Perform one step at a time under teacher or adult guidance where needed.
5. Record date, condition observed, action taken, result, and one improvement.

Rubric: excellent work names the real task, uses safe handling, gives a clear sequence, cares for living things, and records useful evidence.

Animal Breeding Reproductive systems Reproductive stages Breeding systems Selection and culling Methods: reproductive systems of animals: Activity

Activity for Animal Breeding Reproductive systems Reproductive stages Breeding systems Selection and culling Methods: reproductive systems of animals: Complete a supervised agriculture procedure card.

Inquiry question: Why are animals selected for breeding?

1. State the purpose of the task.
2. List exact tools, materials, living things, and protective items needed.
3. Write the steps in order, including what the teacher or adult must supervise.
4. Name at least three hazards and the safety action for each hazard.
5. Record what should be observed before, during, and after the task.
6. Judge quality using practical criteria: safety, hygiene, care for living things, correct method, and useful records.

Success criteria:

- steps are practical
- tools/materials are named
- safety and care are included

Home link: Observe a related farm, garden, kitchen, storage, or animal-care practice. Record what was done safely and one improvement.

Animal Breeding Reproductive systems Reproductive stages Breeding systems Selection and culling Methods: reproductive systems of animals: Activity And Practice

Practice for Animal Breeding Reproductive systems Reproductive stages Breeding systems Selection and culling Methods: reproductive systems of animals:

1. Carry out or describe a practical task for this outcome: describe reproductive systems of animals. List materials, steps, safety checks, and how you will judge quality.
2. Carry out or describe a practical task for this outcome: examine reproductive stages in animal production. List materials, steps, safety checks, and how you will judge quality.
3. Carry out or describe a practical task for this outcome: assess breeding systems in animal production. List materials, steps, safety checks, and how you will judge quality.
4. Carry out or describe a practical task for this outcome: select breeding stock in animal production. List materials, steps, safety checks, and how you will judge quality.

Correction check:

- steps are practical
- tools/materials are named
- safety and care are included

Reflection: What was easy? What needs more practice? What question will you ask your teacher or study partner?

Home link: Ask someone at home for an example related to animal breeding reproductive systems reproductive stages breeding systems selection and culling methods: reproductive systems of animals and write two useful sentences about it.

Animal Breeding Reproductive systems Reproductive stages Breeding systems Selection and culling Methods: reproductive systems of animals: Outcome Check

Outcome: describe reproductive systems of animals

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Animal Breeding Reproductive systems Reproductive stages Breeding systems Selection and culling Methods: reproductive systems of animals.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Animal Breeding Reproductive systems Reproductive stages Breeding systems Selection and culling Methods: reproductive systems of animals: Outcome Check

Outcome: examine reproductive stages in animal production

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Animal Breeding Reproductive systems Reproductive stages Breeding systems Selection and culling Methods: reproductive systems of animals.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Animal Breeding Reproductive systems Reproductive stages Breeding systems Selection and culling Methods: reproductive systems of animals: Outcome Check

Outcome: assess breeding systems in animal production

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Animal Breeding Reproductive systems Reproductive stages Breeding systems Selection and culling Methods: reproductive systems of animals.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Animal Breeding Reproductive systems Reproductive stages Breeding systems Selection and culling Methods: evaluate various methods: Lesson Opener

Learning area: Animal Breeding Reproductive systems Reproductive stages Breeding systems Selection and culling Methods: evaluate various methods

Start here: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Animal Breeding Reproductive systems Reproductive stages Breeding systems Selection and culling Methods: evaluate various methods.

Curriculum link: This lesson follows the Grade 12 curriculum sequence and focuses on Animal Breeding Reproductive systems Reproductive stages Breeding systems Selection and culling Methods: evaluate various methods.

By the end of this lesson sequence, you should be able to:

- evaluate various methods of service in animal breeding
- acknowledge the role of breeding in animal production.

Inquiry questions:

- How are animals selected for breeding?

Before reading, write one thing you already know and one question you want this lesson to answer.

Animal Breeding Reproductive systems Reproductive stages Breeding systems Selection and culling Methods: evaluate various methods: Learn

Start with this real situation: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Animal Breeding Reproductive systems Reproductive stages Breeding systems Selection and culling Methods: evaluate various methods.

The important idea is Animal Breeding Reproductive systems Reproductive stages Breeding systems Selection and culling Methods: evaluate various methods. In this lesson, do not memorise words only. Read the example, try the guided task, then explain the idea in your own words.

A common mistake is describing farm work without naming tools, steps, safety, observations, or care practices. Avoid it by checking your work against these success criteria:

- steps are practical
- tools/materials are named
- safety and care are included

For Grade 12, use exam-ready precision with terms explained before use.

Inline visual support:

| Agriculture record | What to note |

| --- | --- |

| Purpose | Why the task is being done. |

| Tools/materials | Exact items needed for Animal Breeding Reproductive systems Reproductive stages Breeding systems Selection and culling Methods: evaluate various methods. |

| Safety and care | Hazard, hygiene action, and care for living things. |

| Observation | Date, condition seen, action taken, and result. |

Method for Agriculture: Learn through school-garden practice, observation logs, tool safety, and home projects.

Key words:

- Animal Breeding Reproductive systems Reproductive stages Breeding systems Selection and culling Methods: evaluate various methods
- Animal Breeding Reproductive systems Reproductive stages Breeding systems Selection and culling Methods of service Role
- Animal
- Breeding
- Reproductive
- systems
- stages
- Selection

Explain the idea in your own words before attempting the activities.

Animal Breeding Reproductive systems Reproductive stages Breeding systems Selection and culling Methods: evaluate various methods: Worked Example

Practical model for Animal Breeding Reproductive systems Reproductive stages Breeding systems Selection and culling Methods: evaluate various methods: Choose the exact agriculture action, then prove it with evidence.

Procedure:

1. Choose one concrete task from this topic: observe soil, care for a crop, handle a tool, feed or protect an animal, store produce, manage water, or keep a record.
2. State the purpose of that task in one sentence.
3. List exact tools, materials, living things, and safety items before starting.
4. Perform one step at a time under teacher or adult guidance where needed.
5. Record date, condition observed, action taken, result, and one improvement.

Rubric: excellent work names the real task, uses safe handling, gives a clear sequence, cares for living things, and records useful evidence.

Animal Breeding Reproductive systems Reproductive stages Breeding systems Selection and culling Methods: evaluate various methods: Activity

Activity for Animal Breeding Reproductive systems Reproductive stages Breeding systems Selection and culling Methods: evaluate various methods: Complete a supervised agriculture procedure card.

Inquiry question: How are animals selected for breeding?

1. State the purpose of the task.
2. List exact tools, materials, living things, and protective items needed.
3. Write the steps in order, including what the teacher or adult must supervise.
4. Name at least three hazards and the safety action for each hazard.
5. Record what should be observed before, during, and after the task.
6. Judge quality using practical criteria: safety, hygiene, care for living things, correct method, and useful records.

Success criteria:

- steps are practical
- tools/materials are named
- safety and care are included

Home link: Observe a related farm, garden, kitchen, storage, or animal-care practice. Record what was done safely and one improvement.

Animal Breeding Reproductive systems Reproductive stages Breeding systems Selection and culling Methods: evaluate various methods: Activity And Practice

Practice for Animal Breeding Reproductive systems Reproductive stages Breeding systems Selection and culling Methods: evaluate various methods:

1. Carry out or describe a practical task for this outcome: evaluate various methods of service in animal breeding. List materials, steps, safety checks, and how you will judge quality.
2. Carry out or describe a practical task for this outcome: acknowledge the role of breeding in animal production. List materials, steps, safety checks, and how you will judge quality.

Correction check:

- steps are practical
- tools/materials are named
- safety and care are included

Reflection: What was easy? What needs more practice? What question will you ask your teacher or study partner?

Home link: Ask someone at home for an example related to animal breeding reproductive systems reproductive stages breeding systems selection and culling methods: evaluate various methods and write two useful sentences about it.

Animal Breeding Reproductive systems Reproductive stages Breeding systems Selection and culling Methods: evaluate various methods: Outcome Check

Outcome: evaluate various methods of service in animal breeding

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Animal Breeding Reproductive systems Reproductive stages Breeding systems Selection and culling Methods: evaluate various methods.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Animal Breeding Reproductive systems Reproductive stages Breeding systems Selection and culling Methods: evaluate various methods: Outcome Check

Outcome: acknowledge the role of breeding in animal production.

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Animal Breeding Reproductive systems Reproductive stages Breeding systems Selection and culling Methods: evaluate various methods.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Animal Health: Disease Control Predisposing factors Categories of animal diseases Disease control: Lesson Opener

Learning area: Animal Health: Disease Control Predisposing factors Categories of animal diseases Disease control

Start here: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Animal Health: Disease Control Predisposing factors Categories of animal diseases Disease control.

Curriculum link: This lesson follows the Grade 12 curriculum sequence and focuses on Animal Health: Disease Control Predisposing factors Categories of animal diseases Disease control.

By the end of this lesson sequence, you should be able to:

- explain factors that predispose animals to diseases
- examine diseases that affect animal production
- apply control measures against animal diseases
- acknowledge importance of disease control in animal production.

Inquiry questions:

- How does animal health influence productivity?
- How are animal diseases controlled?

Before reading, write one thing you already know and one question you want this lesson to answer.

Animal Health: Disease Control Predisposing factors Categories of animal diseases Disease control: Learn

Start with this real situation: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Animal Health: Disease Control Predisposing factors Categories of animal diseases Disease control.

The important idea is Animal Health: Disease Control Predisposing factors Categories of animal diseases Disease control. In this lesson, do not memorise words only. Read the example, try the guided task, then explain the idea in your own words.

A common mistake is describing farm work without naming tools, steps, safety, observations, or care practices. Avoid it by checking your work against these success criteria:

- steps are practical
- tools/materials are named
- safety and care are included

For Grade 12, use exam-ready precision with terms explained before use.

Inline visual support:

| Agriculture record | What to note |

| --- | --- |

| Purpose | Why the task is being done. |

| Tools/materials | Exact items needed for Animal Health: Disease Control Predisposing factors Categories of animal diseases Disease control. |

| Safety and care | Hazard, hygiene action, and care for living things. |

| Observation | Date, condition seen, action taken, and result. |

Method for Agriculture: Learn through school-garden practice, observation logs, tool safety, and home projects.

Key words:

- Animal Health: Disease Control Predisposing factors Categories of animal diseases Disease control
- Animal Health: Disease Control Predisposing factors Categories of animal diseases Disease control measures Importance of disease
- Animal
- Health
- Disease
- Control
- Predisposing

Explain the idea in your own words before attempting the activities.

Animal Health: Disease Control Predisposing factors Categories of animal diseases Disease control: Worked Example

Practical model for Animal Health: Disease Control Predisposing factors Categories of animal diseases Disease control: Crop-care decision card.

Procedure:

1. Identify the crop stage: seed, nursery, transplanted crop, flowering, fruiting, or harvest.
2. Observe spacing, water, weeds, pests, disease signs, soil cover, and plant colour.
3. Choose one safe action: watering, thinning, weeding, mulching, staking, pest scouting, or harvesting.
4. Record why that action fits the evidence seen.
5. Review the result after a set time and note whether plant health improved.

Hazards: sharp tools, unsafe chemicals, dirty water, plant irritation, and overwatering.

Rubric: excellent work uses real observations, names the crop-care action, protects safety, and explains the expected result.

Animal Health: Disease Control Predisposing factors Categories of animal diseases Disease control: Activity

Activity for Animal Health: Disease Control Predisposing factors Categories of animal diseases Disease control: Complete a supervised agriculture procedure card.

Inquiry question: How does animal health influence productivity?

1. State the purpose of the task.
2. List exact tools, materials, living things, and protective items needed.
3. Write the steps in order, including what the teacher or adult must supervise.
4. Name at least three hazards and the safety action for each hazard.
5. Record what should be observed before, during, and after the task.
6. Judge quality using practical criteria: safety, hygiene, care for living things, correct method, and useful records.

Success criteria:

- steps are practical
- tools/materials are named
- safety and care are included

Home link: Observe a related farm, garden, kitchen, storage, or animal-care practice. Record what was done safely and one improvement.

Animal Health: Disease Control Predisposing factors Categories of animal diseases Disease control: Activity And Practice

Practice for Animal Health: Disease Control Predisposing factors Categories of animal diseases
Disease control:

1. Carry out or describe a practical task for this outcome: explain factors that predispose animals to diseases. List materials, steps, safety checks, and how you will judge quality.
2. Carry out or describe a practical task for this outcome: examine diseases that affect animal production. List materials, steps, safety checks, and how you will judge quality.
3. Carry out or describe a practical task for this outcome: apply control measures against animal diseases. List materials, steps, safety checks, and how you will judge quality.
4. Carry out or describe a practical task for this outcome: acknowledge importance of disease control in animal production. List materials, steps, safety checks, and how you will judge quality.

Correction check:

- steps are practical
- tools/materials are named
- safety and care are included

Reflection: What was easy? What needs more practice? What question will you ask your teacher or study partner?

Home link: Ask someone at home for an example related to animal health: disease control predisposing factors categories of animal diseases disease control and write two useful sentences about it.

Animal Health: Disease Control Predisposing factors Categories of animal diseases Disease control: Outcome Check

Outcome: explain factors that predispose animals to diseases

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Animal Health: Disease Control Predisposing factors Categories of animal diseases Disease control.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Animal Health: Disease Control Predisposing factors Categories of animal diseases Disease control: Outcome Check

Outcome: examine diseases that affect animal production

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Animal Health: Disease Control Predisposing factors Categories of animal diseases Disease control.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Animal Health: Disease Control Predisposing factors Categories of animal diseases Disease control: Outcome Check

Outcome: apply control measures against animal diseases

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Animal Health: Disease Control Predisposing factors Categories of animal diseases Disease control.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Non-conventional Livestock and Companion Animals Non-conventional livestock Companion animals Roles of companion: Lesson Opener

Learning area: Non-conventional Livestock and Companion Animals Non-conventional livestock Companion animals Roles of companion

Start here: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Non-conventional Livestock and Companion Animals Non-conventional livestock Companion animals Roles of companion.

Curriculum link: This lesson follows the Grade 12 curriculum sequence and focuses on Non-conventional Livestock and Companion Animals Non-conventional livestock Companion animals Roles of companion.

By the end of this lesson sequence, you should be able to:

- justify the rearing of non-conventional livestock in the community
- examine the rearing of companion animals in the community
- rear selected animal for companionship
- appreciate the role of companion animals and non-conventional livestock in human livelihood.

Inquiry questions:

- How are non-conventional livestock and companion animals beneficial to humans?

Before reading, write one thing you already know and one question you want this lesson to answer.

Non-conventional Livestock and Companion Animals

Non-conventional livestock Companion animals Roles of companion: Learn

Start with this real situation: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Non-conventional Livestock and Companion Animals
Non-conventional livestock Companion animals Roles of companion.

The important idea is Non-conventional Livestock and Companion Animals Non-conventional livestock Companion animals Roles of companion. In this lesson, do not memorise words only. Read the example, try the guided task, then explain the idea in your own words.

A common mistake is describing farm work without naming tools, steps, safety, observations, or care practices. Avoid it by checking your work against these success criteria:

- steps are practical
- tools/materials are named
- safety and care are included

For Grade 12, use exam-ready precision with terms explained before use.

Inline visual support:

| Agriculture record | What to note |

| --- | --- |

| Purpose | Why the task is being done. |

| Tools/materials | Exact items needed for Non-conventional Livestock and Companion Animals
Non-conventional livestock Companion animals Roles of companion. |

| Safety and care | Hazard, hygiene action, and care for living things. |

| Observation | Date, condition seen, action taken, and result. |

Method for Agriculture: Learn through school-garden practice, observation logs, tool safety, and home projects.

Key words:

- Non-conventional Livestock and Companion Animals Non-conventional livestock Companion animals Roles of companion
- Non-conventional Livestock and Companion Animals Non-conventional livestock Companion animals Roles of companion animals and non-conventional livestock.
- Non-conventional
- Livestock
- Companion
- Animals

Explain the idea in your own words before attempting the activities.

Non-conventional Livestock and Companion Animals

Non-conventional livestock Companion animals Roles of companion: Worked Example

Practical model for Non-conventional Livestock and Companion Animals Non-conventional livestock Companion animals Roles of companion: Safe animal handling.

Teacher-supervised procedure:

1. Observe the animal's behaviour before moving close.
2. Approach from the side where the animal can see you.
3. Keep hands away from the mouth, horns, hooves, and back legs.
4. Use clean water, feed, bedding, and equipment as instructed.
5. Wash hands and disinfect equipment after the task.

Hazards: kicks, bites, scratches, disease transmission, and stress to the animal.

Quality criteria: calm handling, clean tools, correct restraint if needed, animal welfare, and accurate records of feed, health, or behaviour.

Non-conventional Livestock and Companion Animals

Non-conventional livestock Companion animals Roles of companion: Activity

Activity for Non-conventional Livestock and Companion Animals Non-conventional livestock Companion animals Roles of companion: Complete a supervised agriculture procedure card.

Inquiry question: How are non-conventional livestock and companion animals beneficial to humans?

1. State the purpose of the task.
2. List exact tools, materials, living things, and protective items needed.
3. Write the steps in order, including what the teacher or adult must supervise.
4. Name at least three hazards and the safety action for each hazard.
5. Record what should be observed before, during, and after the task.
6. Judge quality using practical criteria: safety, hygiene, care for living things, correct method, and useful records.

Success criteria:

- steps are practical
- tools/materials are named
- safety and care are included

Home link: Observe a related farm, garden, kitchen, storage, or animal-care practice. Record what was done safely and one improvement.

Non-conventional Livestock and Companion Animals

Non-conventional livestock Companion animals Roles of companion: Activity And Practice

Practice for Non-conventional Livestock and Companion Animals Non-conventional livestock
Companion animals Roles of companion:

1. Carry out or describe a practical task for this outcome: justify the rearing of non-conventional livestock in the community. List materials, steps, safety checks, and how you will judge quality.
2. Carry out or describe a practical task for this outcome: examine the rearing of companion animals in the community. List materials, steps, safety checks, and how you will judge quality.
3. Carry out or describe a practical task for this outcome: rear selected animal for companionship. List materials, steps, safety checks, and how you will judge quality.
4. Carry out or describe a practical task for this outcome: appreciate the role of companion animals and non-conventional livestock in human livelihood. List materials, steps, safety checks, and how you will judge quality.

Correction check:

- steps are practical
- tools/materials are named
- safety and care are included

Reflection: What was easy? What needs more practice? What question will you ask your teacher or study partner?

Home link: Ask someone at home for an example related to non-conventional livestock and companion animals non-conventional livestock companion animals roles of companion and write two useful sentences about it.

Non-conventional Livestock and Companion Animals

Non-conventional livestock Companion animals Roles of companion: Outcome Check

Outcome: justify the rearing of non-conventional livestock in the community

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Non-conventional Livestock and Companion Animals Non-conventional livestock Companion animals Roles of companion.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Non-conventional Livestock and Companion Animals

Non-conventional livestock Companion animals Roles of companion: Outcome Check

Outcome: examine the rearing of companion animals in the community

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Non-conventional Livestock and Companion Animals Non-conventional livestock Companion animals Roles of companion.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Non-conventional Livestock and Companion Animals

Non-conventional livestock Companion animals Roles of companion: Outcome Check

Outcome: rear selected animal for companionship

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Non-conventional Livestock and Companion Animals Non-conventional livestock Companion animals Roles of companion.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Animal Production: Chapter Review

Review the chapter carefully.

1. Write five key words from this chapter and explain each one.
2. Choose two units and compare what you learned.
3. Create one poster, chart, dialogue, model, map, or worked solution that teaches this chapter.
4. Answer one inquiry question again. Improve your first answer using what you now know.
5. Rate your confidence: high, medium, or needs practice.

Chapter: Agricultural Technologies and Entrepreneurship

In this chapter you will study Agricultural Technologies and Entrepreneurship.

Inquiry starter: What do you already know about this topic from home, school, or your community?

Chapter units:

- 3.1 Urban and peri-urban Agriculture Structural supports Soil-less media Nutrients and water supply Vertical housing of animal
- 3.2 Powered Agricultural Implements Powered implements Maintenance practices
- 3.3 Agricultural Biotechnology Application of biotechnology Benefits and challenges of biotechnology Legume seed inoculation and yoghurt making.
- 3.4 Agri-business Accounts Balance sheet Profit and loss account.
- 3.5 Business and Career Opportunities in Agriculture Opportunities in Agriculture Risks and uncertainties Agribusiness ideas Agribusiness plan

Before you begin, write two things you already know and one question you want to answer. As you study, collect examples from daily life. At the end of the chapter, return to your question and improve your answer using new vocabulary, examples, and evidence from the lessons.

Urban and peri-urban Agriculture Structural supports Soil-less media Nutrients and water supply: Lesson Opener

Learning area: Urban and peri-urban Agriculture Structural supports Soil-less media Nutrients and water supply

Start here: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Urban and peri-urban Agriculture Structural supports Soil-less media Nutrients and water supply.

Curriculum link: This lesson follows the Grade 12 curriculum sequence and focuses on Urban and peri-urban Agriculture Structural supports Soil-less media Nutrients and water supply.

By the end of this lesson sequence, you should be able to:

- describe agro-technological practices for urban and peri-urban agriculture
- apply selected technologies for urban and peri-urban agriculture
- appraise the practices of urban and peri-urban agriculture in the community.

Inquiry questions:

- What agro-technological practices are suited for urban and peri-urban areas?
- How does urban and peri-urban agriculture contribute to food security?

Before reading, write one thing you already know and one question you want this lesson to answer.

Urban and peri-urban Agriculture Structural supports Soil-less media Nutrients and water supply: Learn

Start with this real situation: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Urban and peri-urban Agriculture Structural supports Soil-less media Nutrients and water supply.

The important idea is Urban and peri-urban Agriculture Structural supports Soil-less media Nutrients and water supply. In this lesson, do not memorise words only. Read the example, try the guided task, then explain the idea in your own words.

A common mistake is describing farm work without naming tools, steps, safety, observations, or care practices. Avoid it by checking your work against these success criteria:

- steps are practical
- tools/materials are named
- safety and care are included

For Grade 12, use exam-ready precision with terms explained before use.

Inline visual support:

| Agriculture record | What to note |

| --- | --- |

| Purpose | Why the task is being done. |

| Tools/materials | Exact items needed for Urban and peri-urban Agriculture Structural supports Soil-less media Nutrients and water supply. |

| Safety and care | Hazard, hygiene action, and care for living things. |

| Observation | Date, condition seen, action taken, and result. |

Method for Agriculture: Learn through school-garden practice, observation logs, tool safety, and home projects.

Key words:

- Urban and peri-urban Agriculture Structural supports Soil-less media Nutrients and water supply
- Urban and peri-urban Agriculture Structural supports Soil-less media Nutrients and water supply
- Vertical housing of animal
- Urban
- peri-urban
- Structural
- supports
- Soil-less
- media

Explain the idea in your own words before attempting the activities.

Urban and peri-urban Agriculture Structural supports Soil-less media Nutrients and water supply: Worked Example

Practical model for Urban and peri-urban Agriculture Structural supports Soil-less media Nutrients and water supply: Soil observation and improvement.

Procedure:

1. Collect a small soil sample from an approved school-garden spot.
2. Observe colour, texture, moisture, stones, plant remains, and signs of erosion.
3. Record the evidence in a table before suggesting an improvement.
4. Choose one improvement such as mulching, compost, contour planting, cover crops, or safe drainage.
5. Explain how the improvement protects soil fertility, water, or plant growth.

Hazards: sharp objects, dirty hands, unsafe slopes, and contaminated soil.

Rubric: excellent work names the soil evidence, links it to a practical improvement, includes safety, and records observations clearly.

Urban and peri-urban Agriculture Structural supports Soil-less media Nutrients and water supply: Activity

Activity for Urban and peri-urban Agriculture Structural supports Soil-less media Nutrients and water supply: Complete a supervised agriculture procedure card.

Inquiry question: What agro-technological practices are suited for urban and peri-urban areas?

1. State the purpose of the task.
2. List exact tools, materials, living things, and protective items needed.
3. Write the steps in order, including what the teacher or adult must supervise.
4. Name at least three hazards and the safety action for each hazard.
5. Record what should be observed before, during, and after the task.
6. Judge quality using practical criteria: safety, hygiene, care for living things, correct method, and useful records.

Success criteria:

- steps are practical
- tools/materials are named
- safety and care are included

Home link: Observe a related farm, garden, kitchen, storage, or animal-care practice. Record what was done safely and one improvement.

Urban and peri-urban Agriculture Structural supports Soil-less media Nutrients and water supply: Activity And Practice

Practice for Urban and peri-urban Agriculture Structural supports Soil-less media Nutrients and water supply:

1. Carry out or describe a practical task for this outcome: describe agro-technological practices for urban and peri-urban agriculture. List materials, steps, safety checks, and how you will judge quality.
2. Carry out or describe a practical task for this outcome: apply selected technologies for urban and peri-urban agriculture. List materials, steps, safety checks, and how you will judge quality.
3. Carry out or describe a practical task for this outcome: appraise the practices of urban and peri-urban agriculture in the community. List materials, steps, safety checks, and how you will judge quality.

Correction check:

- steps are practical
- tools/materials are named
- safety and care are included

Reflection: What was easy? What needs more practice? What question will you ask your teacher or study partner?

Home link: Ask someone at home for an example related to urban and peri-urban agriculture structural supports soil-less media nutrients and water supply and write two useful sentences about it.

Urban and peri-urban Agriculture Structural supports Soil-less media Nutrients and water supply: Outcome Check

Outcome: describe agro-technological practices for urban and peri-urban agriculture

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Urban and peri-urban Agriculture Structural supports Soil-less media Nutrients and water supply.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Urban and peri-urban Agriculture Structural supports Soil-less media Nutrients and water supply: Outcome Check

Outcome: apply selected technologies for urban and peri-urban agriculture

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Urban and peri-urban Agriculture Structural supports Soil-less media Nutrients and water supply.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Urban and peri-urban Agriculture Structural supports Soil-less media Nutrients and water supply: Outcome Check

Outcome: appraise the practices of urban and peri-urban agriculture in the community.

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Urban and peri-urban Agriculture Structural supports Soil-less media Nutrients and water supply.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Powered Agricultural Implements Powered implements Maintenance practices: Lesson Opener

Learning area: Powered Agricultural Implements Powered implements Maintenance practices

Start here: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Powered Agricultural Implements Powered implements Maintenance practices.

Curriculum link: This lesson follows the Grade 12 curriculum sequence and focuses on Powered Agricultural Implements Powered implements Maintenance practices.

By the end of this lesson sequence, you should be able to:

- describe powered implements in agricultural production
- carry out maintenance practices in agricultural powered implements
- appreciate the role of powered implements in agricultural production.

Inquiry questions:

- How do powered implements enhance agricultural production?

Before reading, write one thing you already know and one question you want this lesson to answer.

Powered Agricultural Implements Powered implements Maintenance practices: Learn

Start with this real situation: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Powered Agricultural Implements Powered implements Maintenance practices.

The important idea is Powered Agricultural Implements Powered implements Maintenance practices. In this lesson, do not memorise words only. Read the example, try the guided task, then explain the idea in your own words.

A common mistake is describing farm work without naming tools, steps, safety, observations, or care practices. Avoid it by checking your work against these success criteria:

- steps are practical
- tools/materials are named
- safety and care are included

For Grade 12, use exam-ready precision with terms explained before use.

Inline visual support:

| Agriculture record | What to note |

| --- | --- |

| Purpose | Why the task is being done. |

| Tools/materials | Exact items needed for Powered Agricultural Implements Powered implements Maintenance practices. |

| Safety and care | Hazard, hygiene action, and care for living things. |

| Observation | Date, condition seen, action taken, and result. |

Method for Agriculture: Learn through school-garden practice, observation logs, tool safety, and home projects.

Key words:

- Powered Agricultural Implements Powered implements Maintenance practices
- Powered
- Agricultural
- Implements
- Maintenance
- practices

Explain the idea in your own words before attempting the activities.

Powered Agricultural Implements Powered implements Maintenance practices: Worked Example

Practical model for Powered Agricultural Implements Powered implements Maintenance practices: Tool selection and maintenance check.

Procedure:

1. Name the agriculture task and choose the correct tool for it.
2. Inspect the handle, blade, joint, edge, or moving part before use.
3. Use the tool with safe spacing and the correct body position.
4. Clean, dry, oil, sharpen, or store the tool as appropriate after use.
5. Record one fault found and the safe maintenance action taken.

Hazards: cuts, loose handles, rust, poor storage, and using the wrong tool.

Rubric: excellent work matches tool to task, checks safety before use, maintains the tool correctly, and records the result.

Powered Agricultural Implements Powered implements Maintenance practices: Activity

Activity for Powered Agricultural Implements Powered implements Maintenance practices: Complete a supervised agriculture procedure card.

Inquiry question: How do powered implements enhance agricultural production?

1. State the purpose of the task.
2. List exact tools, materials, living things, and protective items needed.
3. Write the steps in order, including what the teacher or adult must supervise.
4. Name at least three hazards and the safety action for each hazard.
5. Record what should be observed before, during, and after the task.
6. Judge quality using practical criteria: safety, hygiene, care for living things, correct method, and useful records.

Success criteria:

- steps are practical
- tools/materials are named
- safety and care are included

Home link: Observe a related farm, garden, kitchen, storage, or animal-care practice. Record what was done safely and one improvement.

Powered Agricultural Implements Powered implements Maintenance practices: Activity And Practice

Practice for Powered Agricultural Implements Powered implements Maintenance practices:

1. Carry out or describe a practical task for this outcome: describe powered implements in agricultural production. List materials, steps, safety checks, and how you will judge quality.
2. Carry out or describe a practical task for this outcome: carry out maintenance practices in agricultural powered implements. List materials, steps, safety checks, and how you will judge quality.
3. Carry out or describe a practical task for this outcome: appreciate the role of powered implements in agricultural production. List materials, steps, safety checks, and how you will judge quality.

Correction check:

- steps are practical
- tools/materials are named
- safety and care are included

Reflection: What was easy? What needs more practice? What question will you ask your teacher or study partner?

Home link: Ask someone at home for an example related to powered agricultural implements maintenance practices and write two useful sentences about it.

Powered Agricultural Implements Powered implements Maintenance practices: Outcome Check

Outcome: describe powered implements in agricultural production

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Powered Agricultural Implements Powered implements Maintenance practices.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Powered Agricultural Implements Powered implements Maintenance practices: Outcome Check

Outcome: carry out maintenance practices in agricultural powered implements

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Powered Agricultural Implements Maintenance practices.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Powered Agricultural Implements Maintenance practices: Outcome Check

Outcome: appreciate the role of powered implements in agricultural production.

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Powered Agricultural Implements Maintenance practices.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Agricultural Biotechnology Application of biotechnology Benefits and challenges of biotechnology Legume seed: Lesson Opener

Learning area: Agricultural Biotechnology Application of biotechnology Benefits and challenges of biotechnology Legume seed

Start here: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Agricultural Biotechnology Application of biotechnology Benefits and challenges of biotechnology Legume seed.

Curriculum link: This lesson follows the Grade 12 curriculum sequence and focuses on Agricultural Biotechnology Application of biotechnology Benefits and challenges of biotechnology Legume seed.

By the end of this lesson sequence, you should be able to:

- explain application of biotechnology in agricultural production
- assess selected outputs of biotechnology in agriculture
- carry out selected biotechnological practice in agriculture
- appreciate the role of biotechnology in agricultural production.

Inquiry questions:

- How is biotechnology used in agricultural production?

Before reading, write one thing you already know and one question you want this lesson to answer.

Agricultural Biotechnology Application of biotechnology Benefits and challenges of biotechnology Legume seed: Learn

Start with this real situation: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Agricultural Biotechnology Application of biotechnology Benefits and challenges of biotechnology Legume seed.

The important idea is Agricultural Biotechnology Application of biotechnology Benefits and challenges of biotechnology Legume seed. In this lesson, do not memorise words only. Read the example, try the guided task, then explain the idea in your own words.

A common mistake is describing farm work without naming tools, steps, safety, observations, or care practices. Avoid it by checking your work against these success criteria:

- steps are practical
- tools/materials are named
- safety and care are included

For Grade 12, use exam-ready precision with terms explained before use.

Inline visual support:

| Agriculture record | What to note |

| --- | --- |

| Purpose | Why the task is being done. |

| Tools/materials | Exact items needed for Agricultural Biotechnology Application of biotechnology Benefits and challenges of biotechnology Legume seed. |

| Safety and care | Hazard, hygiene action, and care for living things. |

| Observation | Date, condition seen, action taken, and result. |

Method for Agriculture: Learn through school-garden practice, observation logs, tool safety, and home projects.

Key words:

- Agricultural Biotechnology Application of biotechnology Benefits and challenges of biotechnology Legume seed
- Agricultural Biotechnology Application of biotechnology Benefits and challenges of biotechnology Legume seed inoculation and yoghurt making.
- Agricultural
- Biotechnology
- Application
- Benefits
- challenges
- Legume

Explain the idea in your own words before attempting the activities.

Agricultural Biotechnology Application of biotechnology Benefits and challenges of biotechnology Legume seed: Worked Example

Practical model for Agricultural Biotechnology Application of biotechnology Benefits and challenges of biotechnology Legume seed: Crop-care decision card.

Procedure:

1. Identify the crop stage: seed, nursery, transplanted crop, flowering, fruiting, or harvest.
2. Observe spacing, water, weeds, pests, disease signs, soil cover, and plant colour.
3. Choose one safe action: watering, thinning, weeding, mulching, staking, pest scouting, or harvesting.
4. Record why that action fits the evidence seen.
5. Review the result after a set time and note whether plant health improved.

Hazards: sharp tools, unsafe chemicals, dirty water, plant irritation, and overwatering.

Rubric: excellent work uses real observations, names the crop-care action, protects safety, and explains the expected result.

Agricultural Biotechnology Application of biotechnology Benefits and challenges of biotechnology Legume seed: Activity

Activity for Agricultural Biotechnology Application of biotechnology Benefits and challenges of biotechnology Legume seed: Complete a supervised agriculture procedure card.

Inquiry question: How is biotechnology used in agricultural production?

1. State the purpose of the task.
2. List exact tools, materials, living things, and protective items needed.
3. Write the steps in order, including what the teacher or adult must supervise.
4. Name at least three hazards and the safety action for each hazard.
5. Record what should be observed before, during, and after the task.
6. Judge quality using practical criteria: safety, hygiene, care for living things, correct method, and useful records.

Success criteria:

- steps are practical
- tools/materials are named
- safety and care are included

Home link: Observe a related farm, garden, kitchen, storage, or animal-care practice. Record what was done safely and one improvement.

Agricultural Biotechnology Application of biotechnology Benefits and challenges of biotechnology Legume seed: Activity And Practice

Practice for Agricultural Biotechnology Application of biotechnology Benefits and challenges of biotechnology Legume seed:

1. Carry out or describe a practical task for this outcome: explain application of biotechnology in agricultural production. List materials, steps, safety checks, and how you will judge quality.
2. Carry out or describe a practical task for this outcome: assess selected outputs of biotechnology in agriculture. List materials, steps, safety checks, and how you will judge quality.
3. Carry out or describe a practical task for this outcome: carry out selected biotechnological practice in agriculture. List materials, steps, safety checks, and how you will judge quality.
4. Carry out or describe a practical task for this outcome: appreciate the role of biotechnology in agricultural production. List materials, steps, safety checks, and how you will judge quality.

Correction check:

- steps are practical
- tools/materials are named
- safety and care are included

Reflection: What was easy? What needs more practice? What question will you ask your teacher or study partner?

Home link: Ask someone at home for an example related to agricultural biotechnology application of biotechnology benefits and challenges of biotechnology legume seed and write two useful sentences about it.

Agricultural Biotechnology Application of biotechnology Benefits and challenges of biotechnology Legume seed: Outcome Check

Outcome: explain application of biotechnology in agricultural production

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Agricultural Biotechnology Application of biotechnology Benefits and challenges of biotechnology Legume seed.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Agricultural Biotechnology Application of biotechnology Benefits and challenges of biotechnology Legume seed: Outcome Check

Outcome: assess selected outputs of biotechnology in agriculture

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Agricultural Biotechnology Application of biotechnology Benefits and challenges of biotechnology Legume seed.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Agricultural Biotechnology Application of biotechnology Benefits and challenges of biotechnology Legume seed: Outcome Check

Outcome: carry out selected biotechnological practice in agriculture

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Agricultural Biotechnology Application of biotechnology Benefits and challenges of biotechnology Legume seed.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Agri-business Accounts Balance sheet Profit and loss account.: Lesson Opener

Learning area: Agri-business Accounts Balance sheet Profit and loss account.

Start here: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Agri-business Accounts Balance sheet Profit and loss account.

Curriculum link: This lesson follows the Grade 12 curriculum sequence and focuses on Agri-business Accounts Balance sheet Profit and loss account.

By the end of this lesson sequence, you should be able to:

- describe various financial statements in agricultural enterprise
- prepare financial statements for an agricultural enterprise
- acknowledge the use of financial statements in agricultural entrepreneurship.

Inquiry questions:

- How are financial statements prepared in an agricultural enterprise?
- Why are financial statements prepared?

Before reading, write one thing you already know and one question you want this lesson to answer.

Agri-business Accounts Balance sheet Profit and loss account.: Learn

Start with this real situation: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Agri-business Accounts Balance sheet Profit and loss account.

The important idea is Agri-business Accounts Balance sheet Profit and loss account. In this lesson, do not memorise words only. Read the example, try the guided task, then explain the idea in your own words.

A common mistake is describing farm work without naming tools, steps, safety, observations, or care practices. Avoid it by checking your work against these success criteria:

- steps are practical
- tools/materials are named
- safety and care are included

For Grade 12, use exam-ready precision with terms explained before use.

Inline visual support:

| Agriculture record | What to note |

| --- | --- |

| Purpose | Why the task is being done. |

| Tools/materials | Exact items needed for Agri-business Accounts Balance sheet Profit and loss account. |

| Safety and care | Hazard, hygiene action, and care for living things. |

| Observation | Date, condition seen, action taken, and result. |

Method for Agriculture: Learn through school-garden practice, observation logs, tool safety, and home projects.

Key words:

- Agri-business Accounts Balance sheet Profit and loss account.
- Agri-business
- Accounts
- Balance
- sheet
- Profit
- account.

Explain the idea in your own words before attempting the activities.

Agri-business Accounts Balance sheet Profit and loss account.: Worked Example

Practical model for Agri-business Accounts Balance sheet Profit and loss account.: Simple farm-record or enterprise check.

Procedure:

1. Name the activity, such as planting, feeding, harvesting, selling, or storing.
2. Record date, item, quantity, cost, income, loss, or observation.
3. Calculate one useful figure, such as total cost, total sales, profit, loss, or stock remaining.
4. Explain one decision the record supports.
5. Store the record neatly for comparison with the next activity.

Hazards: wrong measurements, missing entries, unsafe storage, and confusing income with profit.

Rubric: excellent work has accurate entries, correct calculation, a realistic decision, and neat records.

Agri-business Accounts Balance sheet Profit and loss account.: Activity

Activity for Agri-business Accounts Balance sheet Profit and loss account.: Complete a supervised agriculture procedure card.

Inquiry question: How are financial statements prepared in an agricultural enterprise?

1. State the purpose of the task.
2. List exact tools, materials, living things, and protective items needed.
3. Write the steps in order, including what the teacher or adult must supervise.
4. Name at least three hazards and the safety action for each hazard.
5. Record what should be observed before, during, and after the task.
6. Judge quality using practical criteria: safety, hygiene, care for living things, correct method, and useful records.

Success criteria:

- steps are practical
- tools/materials are named
- safety and care are included

Home link: Observe a related farm, garden, kitchen, storage, or animal-care practice. Record what was done safely and one improvement.

Agri-business Accounts Balance sheet Profit and loss account.: Activity And Practice

Practice for Agri-business Accounts Balance sheet Profit and loss account.:

1. Carry out or describe a practical task for this outcome: describe various financial statements in agricultural enterprise. List materials, steps, safety checks, and how you will judge quality.
2. Carry out or describe a practical task for this outcome: prepare financial statements for an agricultural enterprise. List materials, steps, safety checks, and how you will judge quality.
3. Carry out or describe a practical task for this outcome: acknowledge the use of financial statements in agricultural entrepreneurship. List materials, steps, safety checks, and how you will judge quality.

Correction check:

- steps are practical
- tools/materials are named
- safety and care are included

Reflection: What was easy? What needs more practice? What question will you ask your teacher or study partner?

Home link: Ask someone at home for an example related to agri-business accounts balance sheet profit and loss account. and write two useful sentences about it.

Agri-business Accounts Balance sheet Profit and loss account.: Outcome Check

Outcome: describe various financial statements in agricultural enterprise

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Agri-business Accounts Balance sheet Profit and loss account.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Agri-business Accounts Balance sheet Profit and loss account.: Outcome Check

Outcome: prepare financial statements for an agricultural enterprise

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Agri-business Accounts Balance sheet Profit and loss account.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Agri-business Accounts Balance sheet Profit and loss account.: Outcome Check

Outcome: acknowledge the use of financial statements in agricultural entrepreneurship.

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Agri-business Accounts Balance sheet Profit and loss account.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Business and Career Opportunities in Agriculture Opportunities in Agriculture Risks and uncertainties: opportunities in agricultural production: Lesson Opener

Learning area: Business and Career Opportunities in Agriculture Opportunities in Agriculture Risks and uncertainties: opportunities in agricultural production

Start here: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Business and Career Opportunities in Agriculture Opportunities in Agriculture Risks and uncertainties: opportunities in agricultural production.

Curriculum link: This lesson follows the Grade 12 curriculum sequence and focuses on Business and Career Opportunities in Agriculture Opportunities in Agriculture Risks and uncertainties: opportunities in agricultural production.

By the end of this lesson sequence, you should be able to:

- analyse opportunities in agricultural production
- examine risks and uncertainties in agricultural enterprises
- assess agribusiness ideas in agricultural production
- prepare an agri-business plan for selected business idea

Inquiry questions:

- How does agriculture provide business and career opportunities?

Before reading, write one thing you already know and one question you want this lesson to answer.

Business and Career Opportunities in Agriculture Opportunities in Agriculture Risks and uncertainties: opportunities in agricultural production: Learn

Start with this real situation: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Business and Career Opportunities in Agriculture Opportunities in Agriculture Risks and uncertainties: opportunities in agricultural production.

The important idea is Business and Career Opportunities in Agriculture Opportunities in Agriculture Risks and uncertainties: opportunities in agricultural production. In this lesson, do not memorise words only. Read the example, try the guided task, then explain the idea in your own words.

A common mistake is describing farm work without naming tools, steps, safety, observations, or care practices. Avoid it by checking your work against these success criteria:

- steps are practical
- tools/materials are named
- safety and care are included

For Grade 12, use exam-ready precision with terms explained before use.

Inline visual support:

Agriculture record	What to note
Purpose	Why the task is being done.
Tools/materials	Exact items needed for Business and Career Opportunities in Agriculture Opportunities in Agriculture Risks and uncertainties: opportunities in agricultural production.
Safety and care	Hazard, hygiene action, and care for living things.
Observation	Date, condition seen, action taken, and result.

Method for Agriculture: Learn through school-garden practice, observation logs, tool safety, and home projects.

Key words:

- Business and Career Opportunities in Agriculture Opportunities in Agriculture Risks and uncertainties: opportunities in agricultural production
- Business and Career Opportunities in Agriculture Opportunities in Agriculture Risks and uncertainties Agribusiness ideas Agribusiness plan
- Business
- Career
- Opportunities
- Risks
- uncertainties
- agricultural

Explain the idea in your own words before attempting the activities.

Business and Career Opportunities in Agriculture Opportunities in Agriculture Risks and uncertainties: opportunities in agricultural production: Worked Example

Practical model for Business and Career Opportunities in Agriculture Opportunities in Agriculture Risks and uncertainties: opportunities in agricultural production: Choose the exact agriculture action, then prove it with evidence.

Procedure:

1. Choose one concrete task from this topic: observe soil, care for a crop, handle a tool, feed or protect an animal, store produce, manage water, or keep a record.
2. State the purpose of that task in one sentence.
3. List exact tools, materials, living things, and safety items before starting.
4. Perform one step at a time under teacher or adult guidance where needed.
5. Record date, condition observed, action taken, result, and one improvement.

Rubric: excellent work names the real task, uses safe handling, gives a clear sequence, cares for living things, and records useful evidence.

Business and Career Opportunities in Agriculture Opportunities in Agriculture Risks and uncertainties: opportunities in agricultural production: Activity

Activity for Business and Career Opportunities in Agriculture Opportunities in Agriculture Risks and uncertainties: opportunities in agricultural production: Complete a supervised agriculture procedure card.

Inquiry question: How does agriculture provide business and career opportunities?

1. State the purpose of the task.
2. List exact tools, materials, living things, and protective items needed.
3. Write the steps in order, including what the teacher or adult must supervise.
4. Name at least three hazards and the safety action for each hazard.
5. Record what should be observed before, during, and after the task.
6. Judge quality using practical criteria: safety, hygiene, care for living things, correct method, and useful records.

Success criteria:

- steps are practical
- tools/materials are named
- safety and care are included

Home link: Observe a related farm, garden, kitchen, storage, or animal-care practice. Record what was done safely and one improvement.

Business and Career Opportunities in Agriculture Opportunities in Agriculture Risks and uncertainties: opportunities in agricultural production: Activity And Practice

Practice for Business and Career Opportunities in Agriculture Opportunities in Agriculture Risks and uncertainties: opportunities in agricultural production:

1. Carry out or describe a practical task for this outcome: analyse opportunities in agricultural production. List materials, steps, safety checks, and how you will judge quality.
2. Carry out or describe a practical task for this outcome: examine risks and uncertainties in agricultural enterprises. List materials, steps, safety checks, and how you will judge quality.
3. Carry out or describe a practical task for this outcome: assess agribusiness ideas in agricultural production. List materials, steps, safety checks, and how you will judge quality.
4. Carry out or describe a practical task for this outcome: prepare an agri-business plan for selected business idea. List materials, steps, safety checks, and how you will judge quality.

Correction check:

- steps are practical
- tools/materials are named
- safety and care are included

Reflection: What was easy? What needs more practice? What question will you ask your teacher or study partner?

Home link: Ask someone at home for an example related to business and career opportunities in agriculture opportunities in agriculture risks and uncertainties: opportunities in agricultural production and write two useful sentences about it.

Business and Career Opportunities in Agriculture Opportunities in Agriculture Risks and uncertainties: opportunities in agricultural production: Outcome Check

Outcome: analyse opportunities in agricultural production

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Business and Career Opportunities in Agriculture Opportunities in Agriculture Risks and uncertainties: opportunities in agricultural production.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Business and Career Opportunities in Agriculture Opportunities in Agriculture Risks and uncertainties: opportunities in agricultural production: Outcome Check

Outcome: examine risks and uncertainties in agricultural enterprises

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Business and Career Opportunities in Agriculture Opportunities in Agriculture Risks and uncertainties: opportunities in agricultural production.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Business and Career Opportunities in Agriculture Opportunities in Agriculture Risks and uncertainties: opportunities in agricultural production: Outcome Check

Outcome: assess agribusiness ideas in agricultural production

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Business and Career Opportunities in Agriculture Opportunities in Agriculture Risks and uncertainties: opportunities in agricultural production.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Business and Career Opportunities in Agriculture Opportunities in Agriculture Risks and uncertainties: acknowledge business and career: Lesson Opener

Learning area: Business and Career Opportunities in Agriculture Opportunities in Agriculture Risks and uncertainties: acknowledge business and career

Start here: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Business and Career Opportunities in Agriculture Opportunities in Agriculture Risks and uncertainties: acknowledge business and career.

Curriculum link: This lesson follows the Grade 12 curriculum sequence and focuses on Business and Career Opportunities in Agriculture Opportunities in Agriculture Risks and uncertainties: acknowledge business and career.

By the end of this lesson sequence, you should be able to:

- acknowledge business and career opportunities in agriculture.

Inquiry questions:

- How does agriculture provide business and career opportunities?

Before reading, write one thing you already know and one question you want this lesson to answer.

Business and Career Opportunities in Agriculture Opportunities in Agriculture Risks and uncertainties: acknowledge business and career: Learn

Start with this real situation: a school garden, home garden, farm tool, soil, water, crop, animal-care, or food-security example linked to Business and Career Opportunities in Agriculture Opportunities in Agriculture Risks and uncertainties: acknowledge business and career.

The important idea is Business and Career Opportunities in Agriculture Opportunities in Agriculture Risks and uncertainties: acknowledge business and career. In this lesson, do not memorise words only. Read the example, try the guided task, then explain the idea in your own words.

A common mistake is describing farm work without naming tools, steps, safety, observations, or care practices. Avoid it by checking your work against these success criteria:

- steps are practical
- tools/materials are named
- safety and care are included

For Grade 12, use exam-ready precision with terms explained before use.

Inline visual support:

| Agriculture record | What to note |

| --- | --- |

| Purpose | Why the task is being done. |

| Tools/materials | Exact items needed for Business and Career Opportunities in Agriculture Opportunities in Agriculture Risks and uncertainties: acknowledge business and career. |

| Safety and care | Hazard, hygiene action, and care for living things. |

| Observation | Date, condition seen, action taken, and result. |

Method for Agriculture: Learn through school-garden practice, observation logs, tool safety, and home projects.

Key words:

- Business and Career Opportunities in Agriculture Opportunities in Agriculture Risks and uncertainties: acknowledge business and career
- Business and Career Opportunities in Agriculture Opportunities in Agriculture Risks and uncertainties Agribusiness ideas Agribusiness plan
- Business
- Career
- Opportunities
- Risks
- uncertainties
- acknowledge

Explain the idea in your own words before attempting the activities.

Business and Career Opportunities in Agriculture Opportunities in Agriculture Risks and uncertainties: acknowledge business and career: Worked Example

Practical model for Business and Career Opportunities in Agriculture Opportunities in Agriculture Risks and uncertainties: acknowledge business and career: Choose the exact agriculture action, then prove it with evidence.

Procedure:

1. Choose one concrete task from this topic: observe soil, care for a crop, handle a tool, feed or protect an animal, store produce, manage water, or keep a record.
2. State the purpose of that task in one sentence.
3. List exact tools, materials, living things, and safety items before starting.
4. Perform one step at a time under teacher or adult guidance where needed.
5. Record date, condition observed, action taken, result, and one improvement.

Rubric: excellent work names the real task, uses safe handling, gives a clear sequence, cares for living things, and records useful evidence.

Business and Career Opportunities in Agriculture Opportunities in Agriculture Risks and uncertainties: acknowledge business and career: Activity

Activity for Business and Career Opportunities in Agriculture Opportunities in Agriculture Risks and uncertainties: acknowledge business and career: Complete a supervised agriculture procedure card.

Inquiry question: How does agriculture provide business and career opportunities?

1. State the purpose of the task.
2. List exact tools, materials, living things, and protective items needed.
3. Write the steps in order, including what the teacher or adult must supervise.
4. Name at least three hazards and the safety action for each hazard.
5. Record what should be observed before, during, and after the task.
6. Judge quality using practical criteria: safety, hygiene, care for living things, correct method, and useful records.

Success criteria:

- steps are practical
- tools/materials are named
- safety and care are included

Home link: Observe a related farm, garden, kitchen, storage, or animal-care practice. Record what was done safely and one improvement.

Business and Career Opportunities in Agriculture Opportunities in Agriculture Risks and uncertainties: acknowledge business and career: Activity And Practice

Practice for Business and Career Opportunities in Agriculture Opportunities in Agriculture Risks and uncertainties: acknowledge business and career:

1. Carry out or describe a practical task for this outcome: acknowledge business and career opportunities in agriculture. List materials, steps, safety checks, and how you will judge quality.

Correction check:

- steps are practical
- tools/materials are named
- safety and care are included

Reflection: What was easy? What needs more practice? What question will you ask your teacher or study partner?

Home link: Ask someone at home for an example related to business and career opportunities in agriculture opportunities in agriculture risks and uncertainties: acknowledge business and career and write two useful sentences about it.

Business and Career Opportunities in Agriculture Opportunities in Agriculture Risks and uncertainties: acknowledge business and career: Outcome Check

Outcome: acknowledge business and career opportunities in agriculture.

Show mastery with evidence.

1. Explain the outcome in your own words.
2. Give one local, school, home, community, practical, or text-based example.
3. Complete a short task that proves you can apply the idea in Business and Career Opportunities in Agriculture Opportunities in Agriculture Risks and uncertainties: acknowledge business and career.
4. Use correct vocabulary and include a reason, observation, example, step, or result.
5. Write one mistake a learner might make and how to avoid it.
6. Ask a peer, teacher, or parent to check whether your answer is clear.

Evidence of mastery: your work answers the exact outcome, gives a relevant example, and shows a correction after feedback.

Agricultural Technologies and Entrepreneurship: Chapter Review

Review the chapter carefully.

1. Write five key words from this chapter and explain each one.
2. Choose two units and compare what you learned.
3. Create one poster, chart, dialogue, model, map, or worked solution that teaches this chapter.
4. Answer one inquiry question again. Improve your first answer using what you now know.
5. Rate your confidence: high, medium, or needs practice.

Glossary

Use this glossary to revise important words in Agriculture.

- Soil Erosion Control Cut-off drains Terraces Gabions: A practical agriculture idea that needs tools or materials, safe steps, care for living things, observations, and records.
- Crop Production: A practical agriculture idea that needs tools or materials, safe steps, care for living things, observations, and records.
- Forage Crops Types of forage Growing forage Ensiling forage: A practical agriculture idea that needs tools or materials, safe steps, care for living things, observations, and records.
- Animal Production: A practical agriculture idea that needs tools or materials, safe steps, care for living things, observations, and records.
- Agricultural Technologies and Entrepreneurship: A practical agriculture idea that needs tools or materials, safe steps, care for living things, observations, and records.
- Powered Agricultural Implements Powered implements Maintenance practices: A practical agriculture idea that needs tools or materials, safe steps, care for living things, observations, and records.
- Agri-business Accounts Balance sheet Profit and loss account.: A practical agriculture idea that needs tools or materials, safe steps, care for living things, observations, and records.

Add five more words from your lessons, then write a meaning and one correct example sentence or worked example for each word.

Answer And Teacher Notes

Most questions in this book are open practice tasks. A good answer should:

- respond to the exact question asked;
- use correct vocabulary from Agriculture;
- include a clear example, reason, working, observation, or evidence;
- be neat enough for another learner to follow;
- show correction after feedback.

For calculations, check each step. For language work, check spelling, punctuation, grammar, and meaning. For practical subjects, check safety, materials, observations, and conclusion.

Answer-key guide: Social Studies answers should name the place, people, evidence, meaning, and responsible action. Agriculture answers should name tools, sequence, safety, observation, and care for living things. Creative Arts answers should show planning, technique, safety, improvement, and explanation of choices.

Rubric for self-checking:

- 4: The answer addresses the exact question, uses subject vocabulary correctly, gives evidence or working, and includes a useful check or correction.
- 3: The main answer is correct but one part needs stronger evidence, clearer working, better labels, or a fuller explanation.
- 2: Some understanding is shown, but the answer is incomplete, too general, or hard to follow.
- 1: The answer is guessed, off topic, or missing the evidence needed for the task.

Correction routine: reread the command word, underline the key data or evidence, improve one weak step, and write one sentence explaining why the final answer is reasonable.

Final Project

Create a final project for Agriculture.

Your project must include:

1. A clear title.
2. A real-life problem or question.
3. Evidence from at least three lessons.
4. A drawing, table, map, chart, model, dialogue, performance plan, artwork note, or worked solution.
5. A short reflection explaining what you learned.

Present your project to a classmate, teacher, parent, or study group.